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Report Date: Week of August 31

Work Completion Conflicts: PM short day (professional development), although that was for everyone; I also missed the majority (more than half) of Wednesday due to finishing a test.

Completed Tasks

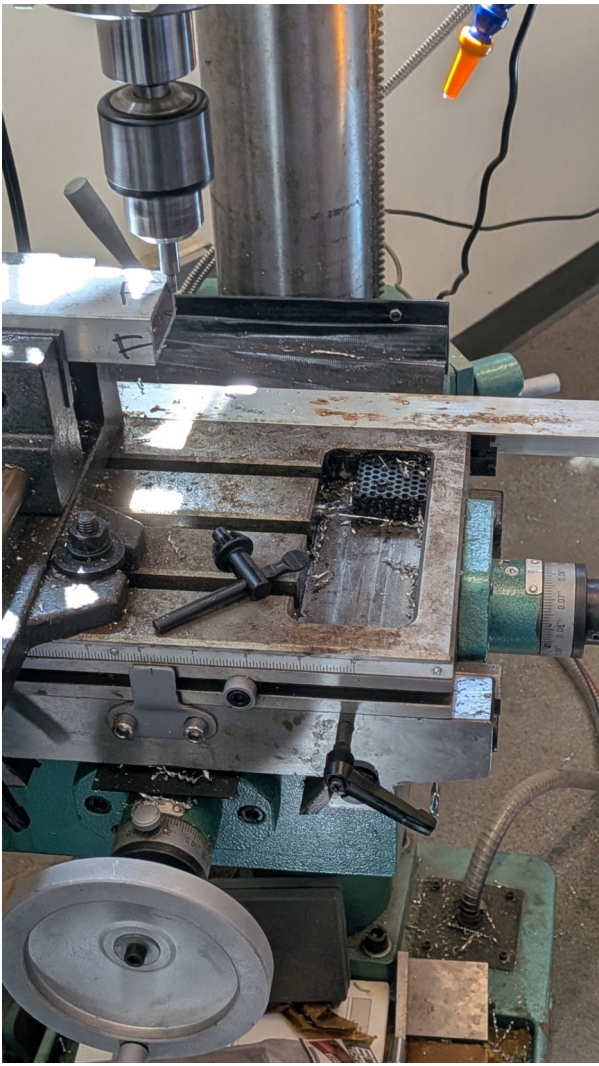
Practicing Drill Pressing - Monday + End of Wednesday - 1 hour 20 minutes (note that I missed most of wednesday, hence the strange timing)

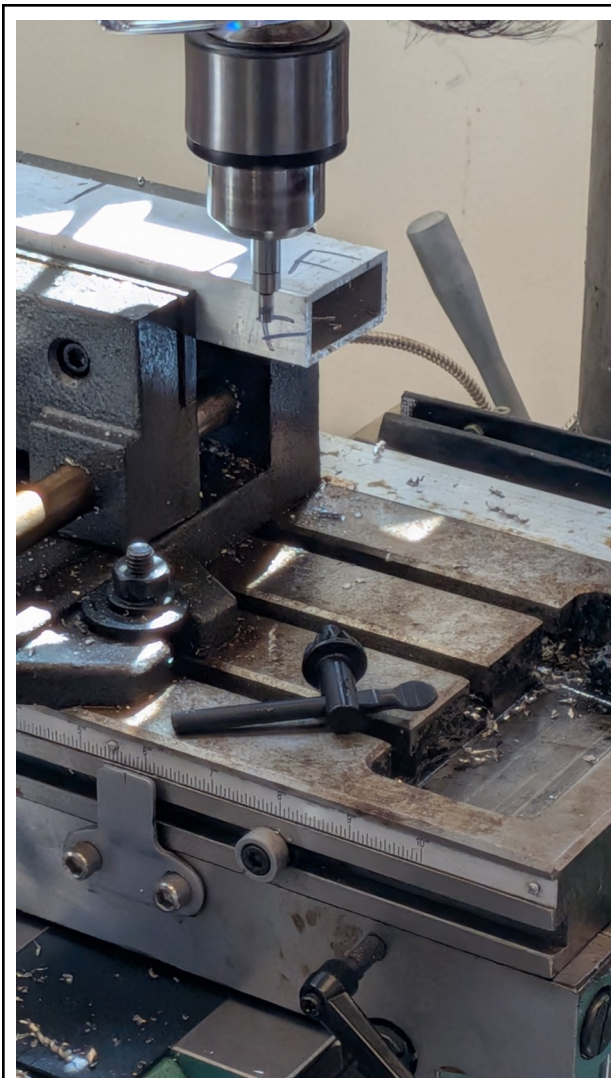


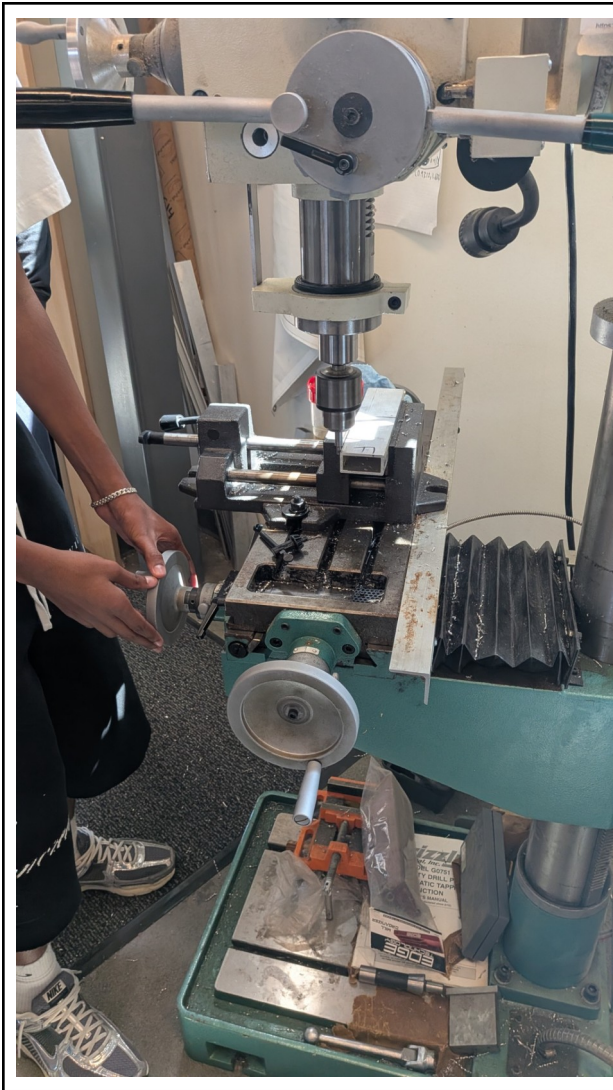
This task focused on building comfort and precision while using the drill press. We used a rectangular prism metal part, as seen in many of the images, as a piece of practicing stock. Before drilling, we had to do a number of preliminary steps. One key part of this was zeroing the drill press using the zeroing bit, something I was not familiar with. It sets a reliable reference point on the metal workpiece, which is important so that the hole position can be measured accurately, from an origin or starting point, rather than being eyeballed or guessed. This step is very important to do correctly because any error in this carries through to literally every hole made after, so we made sure to re-check the zero each time we repositioned the block. Next, I remembered to apply oil, a small/mundane step but still very important. Once zeroed, I drilled a series of holes into the block, paying attention to keep the bit centered and avoiding drifting off. On my first couple attempts, what I noticed is that the bit position would move slightly after touching the metal; I think I was going too fast— this error went away once I started moving a bit slower. I also practiced clearing chips periodically when needed, however, the piece of metal was relatively thin so this



was not a huge issue. However, I know that in the future there are thicker pieces of metal that might necessitate this. Wednesday's shorter session was a continuation of working on the same block. Because I missed a lot of class due to a test, I just had time to complete drilling the remaining holes I was supposed to (for a total of 6), which allowed the limited time I had to still be quite productive.













VIDEO #1:

<https://drive.google.com/file/d/1WB9E-GqLRltUpK48fKfNp3n-y69yRA5c/view?usp=sharing>

You might have to

*download it, the
preview may not
work.*



VIDEO #2:

https://drive.google.com/file/d/1pIO0MPo-h_R87BbGoaLUJzHu8ms0Qipo/view?

usp=sharing

You might have to download it, the preview may not work.

Using the Lathes



This task moved from using the drill press which I was using on Monday and Wednesday to using the Lathe. This was a new use of the lathe, and indeed a different lathe itself, than the ones I had used in Robotics I. Specifically, we drilled into the end of a hex shaft and then threaded the hole we created so that it could accept a screw. We first set up the hex shaft in the Lathe. There was less set up than the drill press, since we did not have to “zero” the lathe in the same way– we were simply drilling down the center. That being said, we still had to oil the piece, and ensure we didn’t leave the chuck key in, a common error that can cause damage and hurt others. Then, I drilled a hole down the center of one end, working carefully and moving slowly. We ensured that the shaft was properly secured before drilling, since a loose workpiece can begin to spin, and that is a significant safety issue.

After the hole was drilled, I used a tap to cut threads into the inside of the hole. Tapping was a much slower, and indeed more “deliberate” process than drilling. It felt this way because the tap has to



VIDEO #3:

<https://drive.google.com/file/d/1AZOnF6ni4dRHtSk-b6l2Gb9xBhdSJtW/view?usp=sharing>

You might have to download it, the preview may not work.

be turned in gradually and backed off periodically to clear metal shavings and prevent damage/binding/snapping to the tap inside the hole, which would ruin the piece and the tap. I would turn the tap a full rotation forward, and then half a rotation backwards, repeating until the threads reached my desired depth. This was my first ever time tapping a hole, so I went slowly and checked alignment often to make sure I wasn't threading at a wrong angle.

Once the tapping was finished, I tested the hole by threading a screw partway into the piece, by hand, to confirm the threads had been cut properly, which I was very glad to see they were!

P.S.-- My phone had hardware damage and was being repaired on Thursday, so I had to request others to film for me. This is why I have considerably less pictures than on Monday/Wednesday; I do apologize, although it was a bit out of my control.